Update of the Indian Ocean albacore Operating Model

10th IOTC Working Party on Methods

14-15 October 2020, Iago MOSQUEIRA

Wageningen Marine Research (WMR), IJmuiden. The Netherlands.

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100years 1918 — 2018

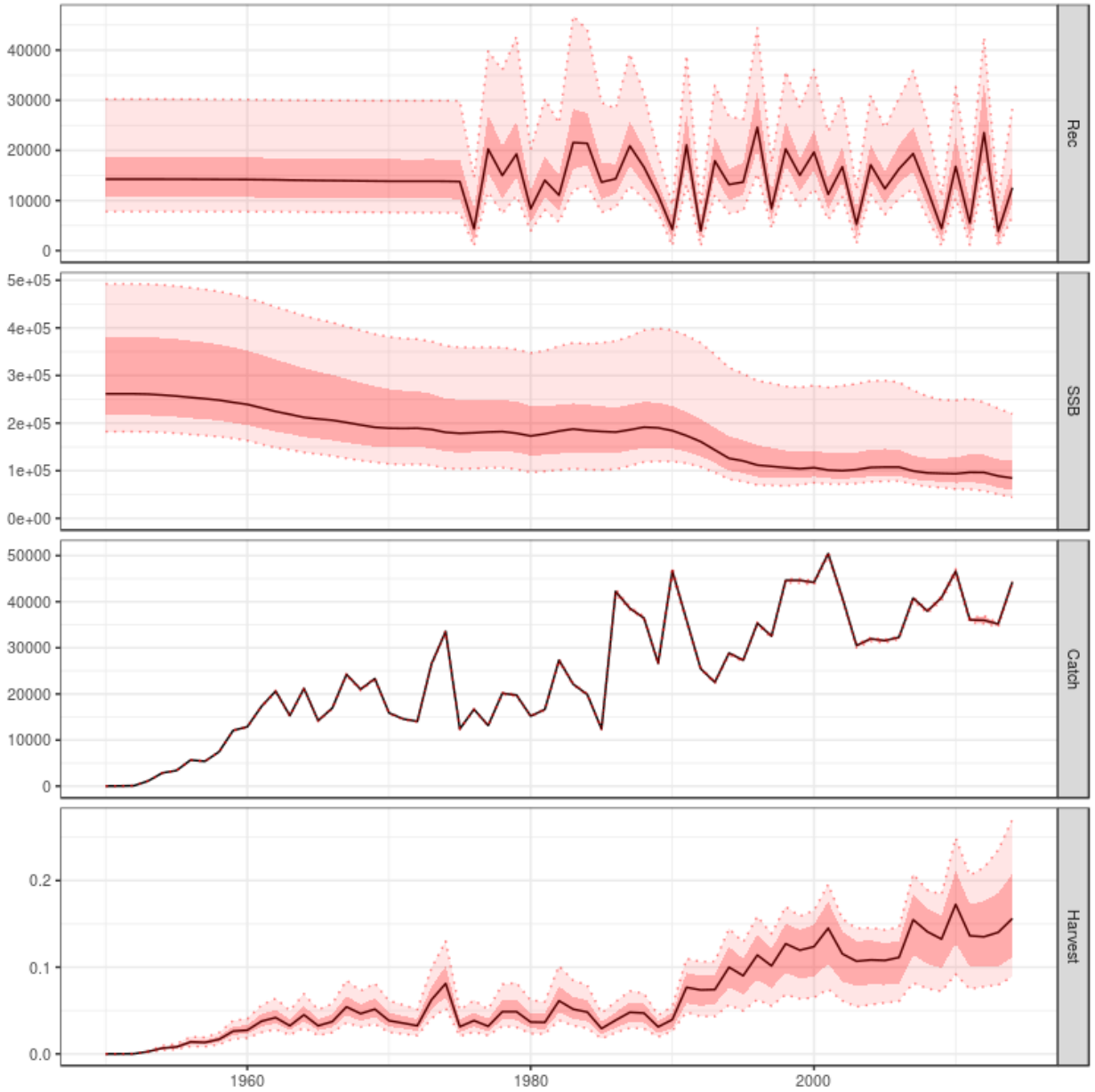
Background

- 3rd iteration of albacore OM.
- To be based on WPTmT 2019 SS3 model.
- Contract WMR - IOTC/FAO signed August.
- Initial exploration new OM.
- Existing FLR platform for SS 3.30.16.
- `https://github.com/iotcwpm/alb`

WPTmT 2014 ALB OM grid

Factor	N	Prod	Values
Natural mortality	5	5	0202, 0303, 0404, 0403, 0402
Steepness SRR	3	15	0.7, 0.8, 0.9
sigma reecruitment	2	30	0.4, 0.6
ESS length comps.	3	90	20, 50, 100
CV CPUE	4	360	0.2, 0.3, 0.4, 0.5
LL q increase	2	720	0%, 0.25% per quarter
Selectivity form	2	1440	logistic, double normal

WPTmT 2014 ALB OM



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100years
1918 — 2018

The WPTmT 2019 SS3 albacore stock assessment

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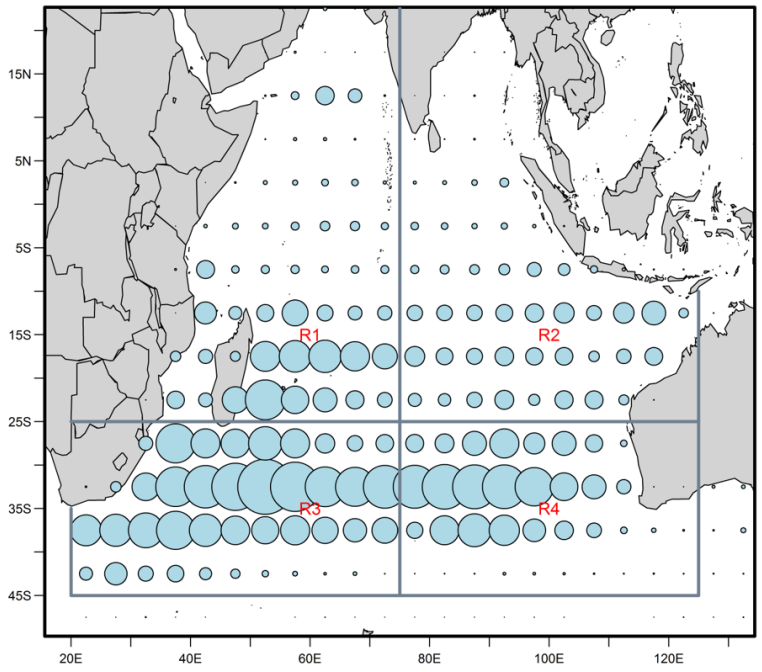
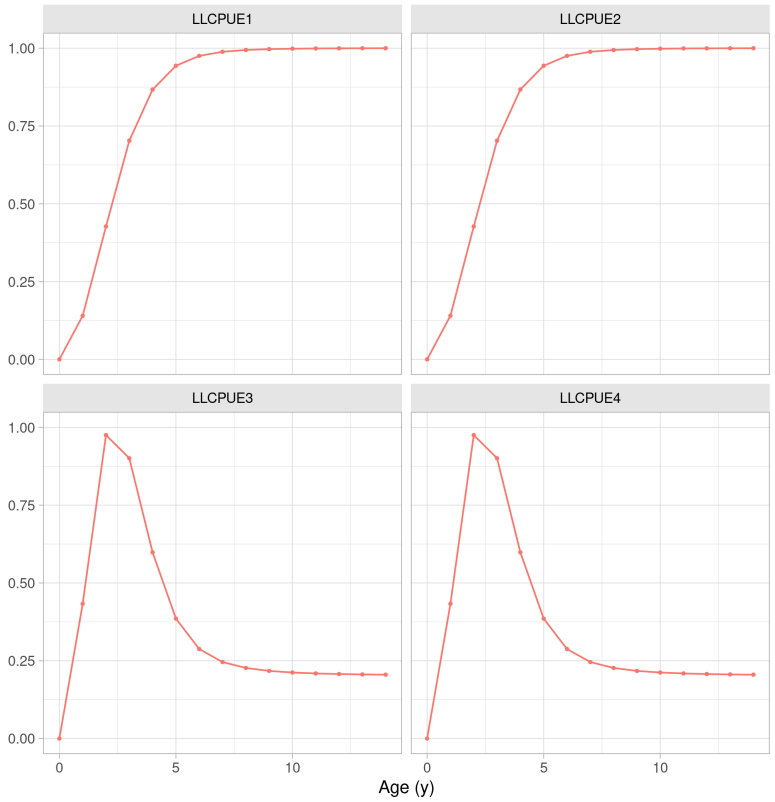
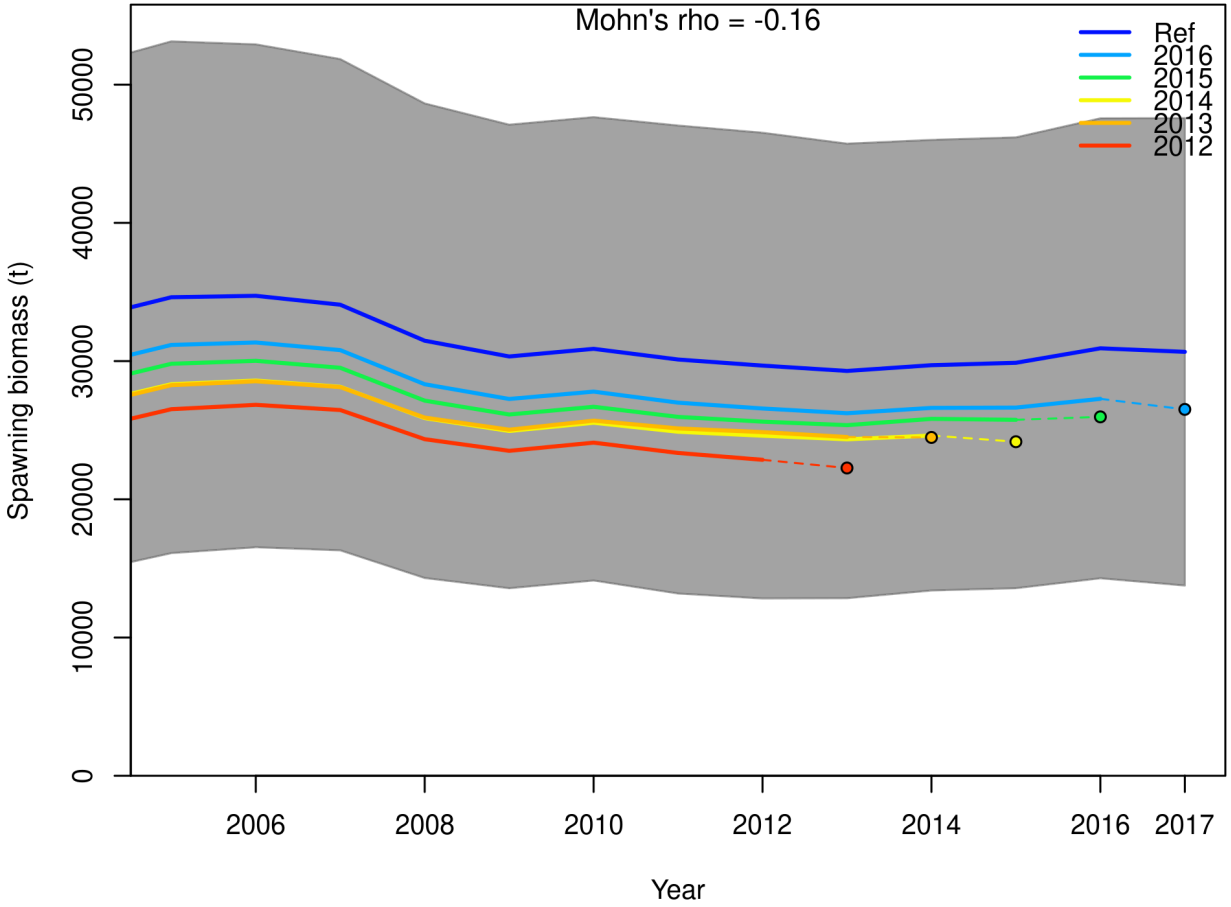
100years 1918 — 2018

Fleet spatial structure

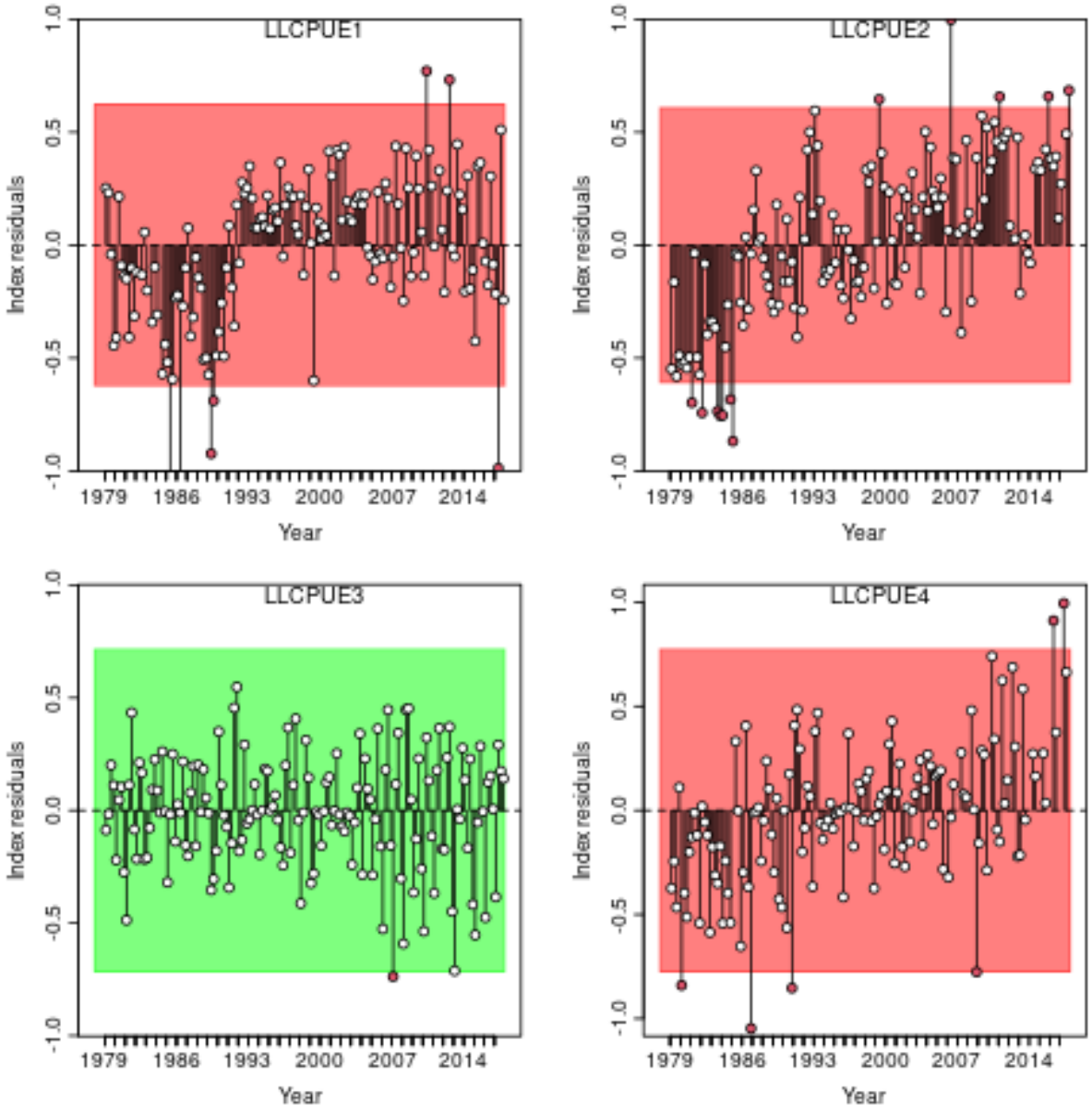




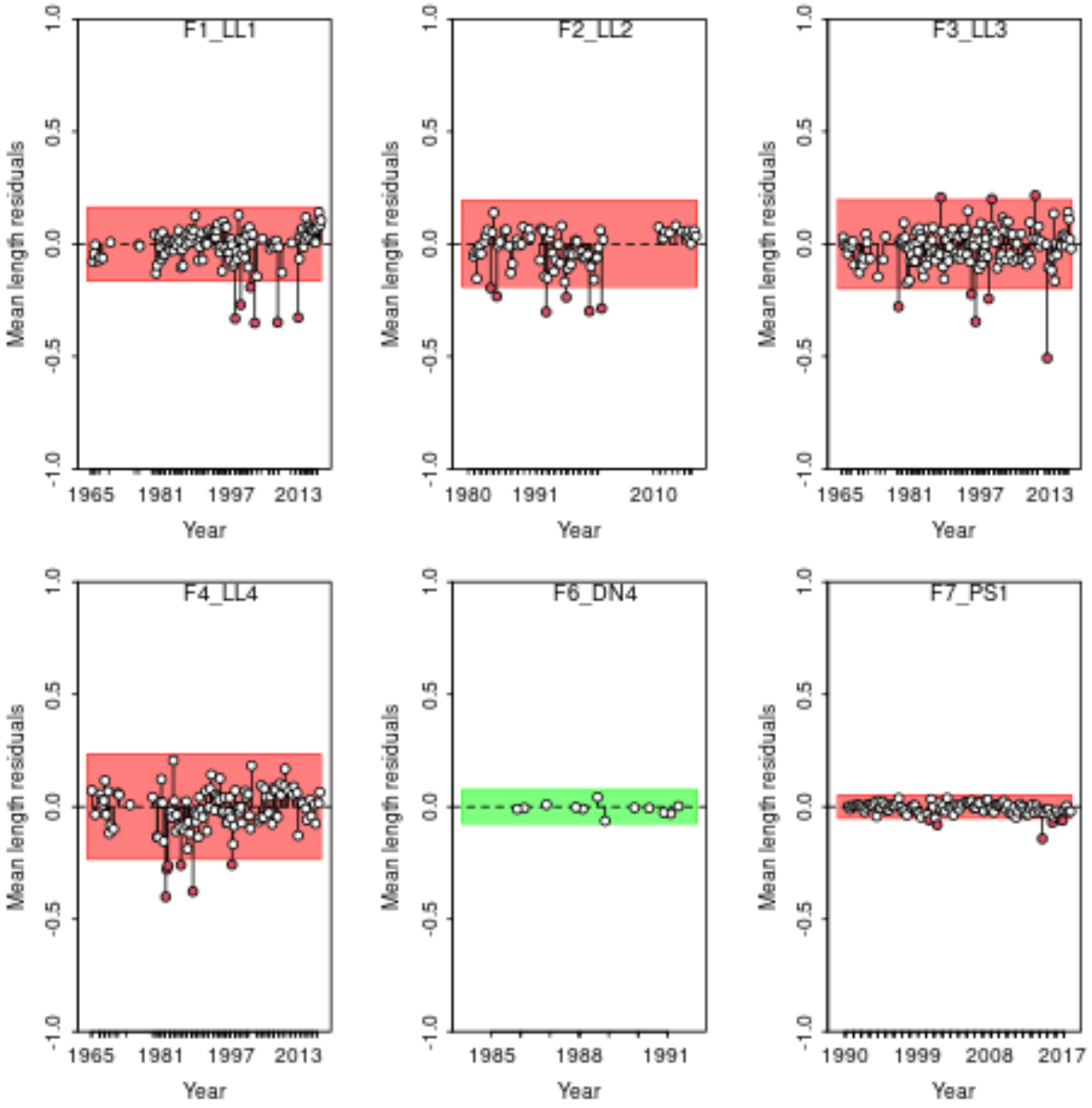
Model diagnostics: Retrospective analysis



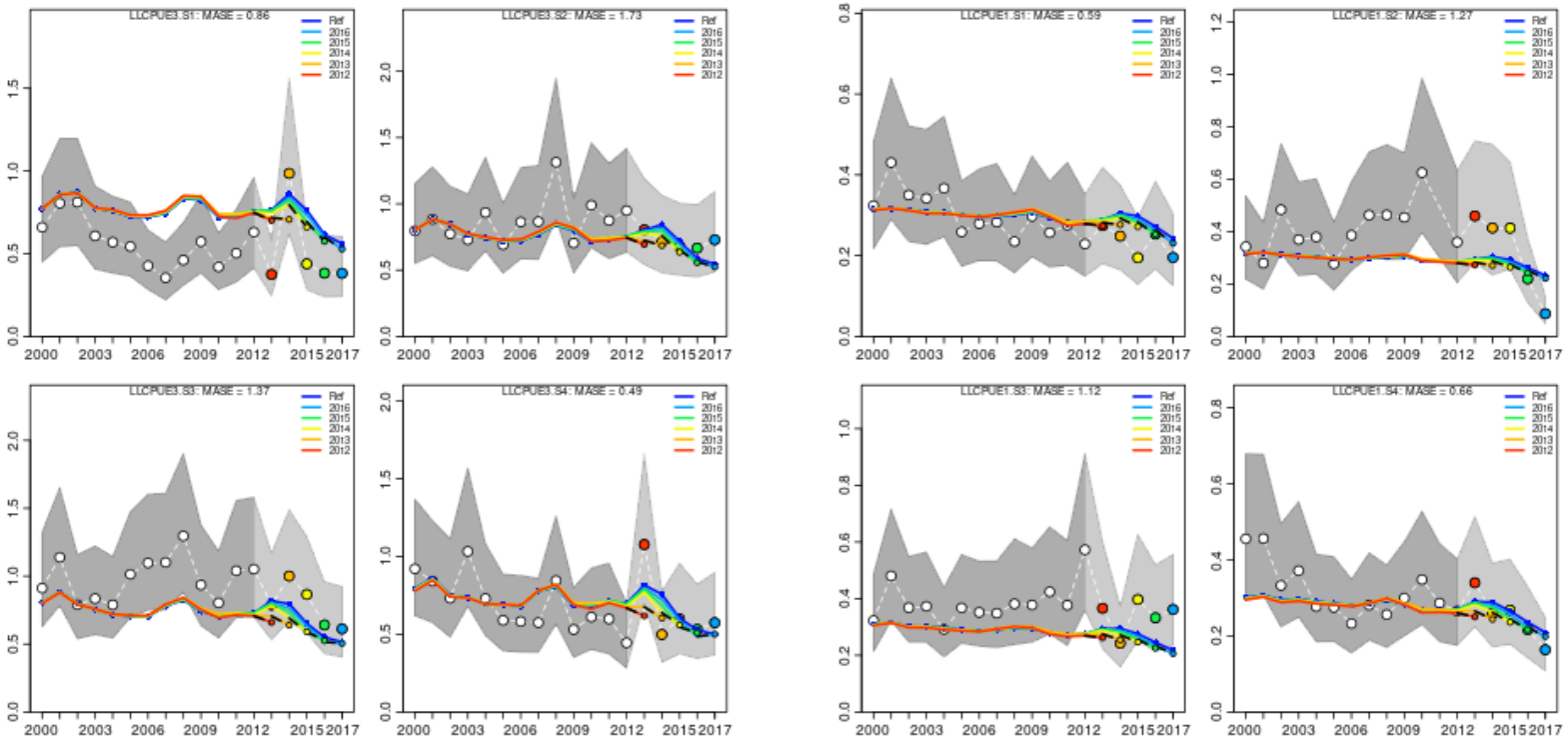
Model diagnostics: CPUE runs tests



Model diagnostics: LF runs tests

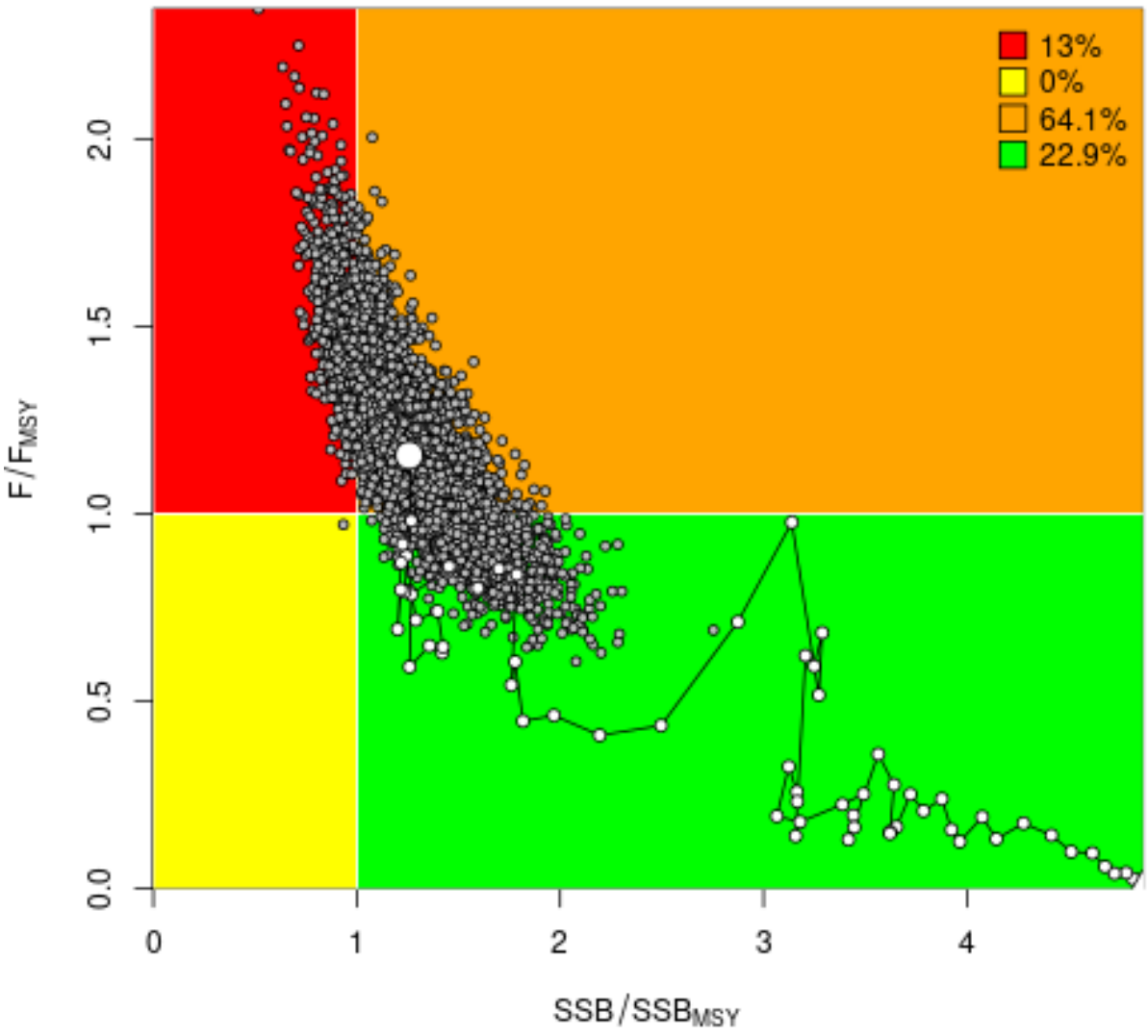


Model diagnostics: Hindcasting cross validation



- MASE < 1: LLCPUE3 S4, LLCPUE1 S1, S4

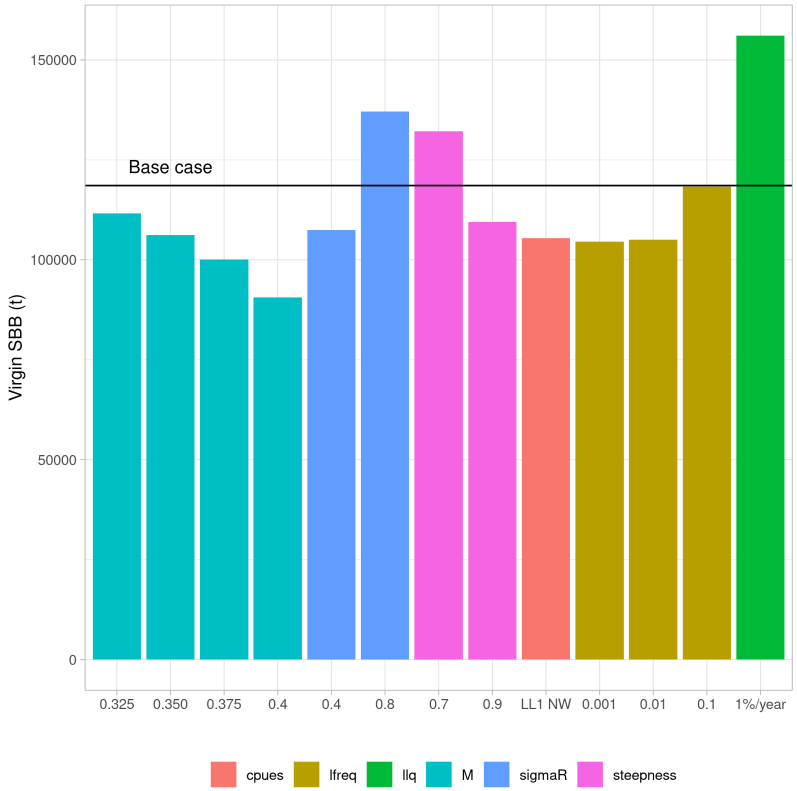
Parameter uncertainty: MVLN

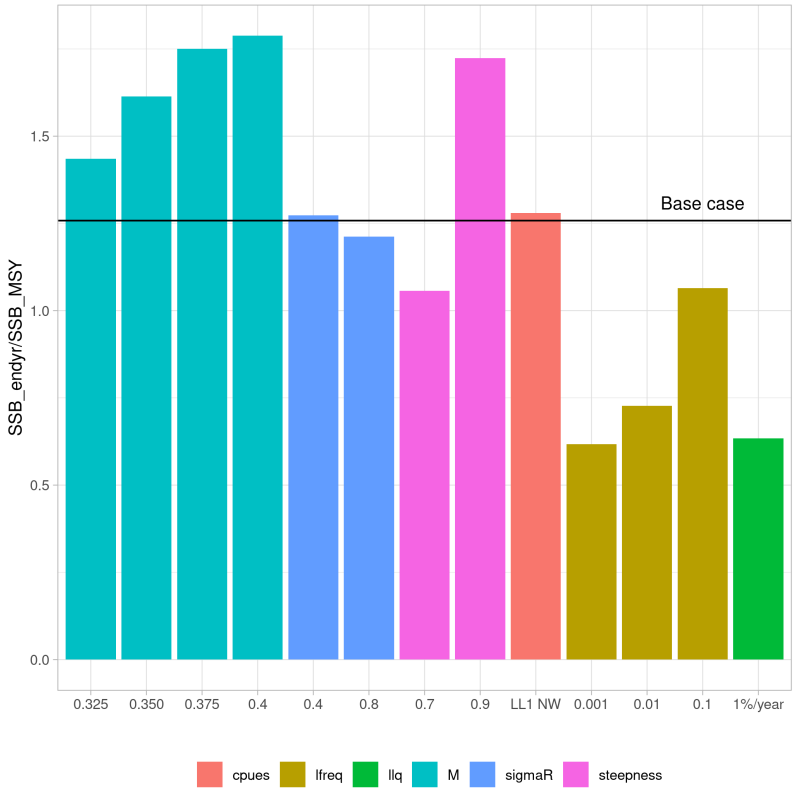


Initial Operating Model grid

- Natural mortality (M): 0.3, 0.325, 0.35, 0.375 or 0.4, for all ages.
- SD recruitment deviates (sigmaR): 0.4, 0.6, or 0.8.
- SRR steepness (h): 0.7, 0.8 or 0.9.
- LL CPUE series (cpues): Northwest (12) or Southwest (14).
- LF data lkhd weighting (lfreq): 0.001, 0.01, 0.1 or 1.
- Catchability increase LL CPUE (llq): 0% or 1% per year.

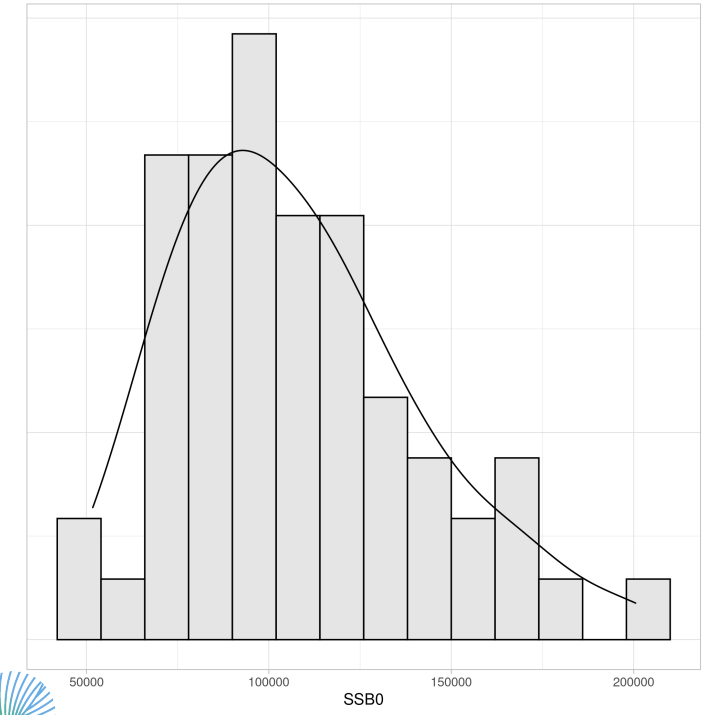
Main effects: Change in SB0, SBMSY by factor



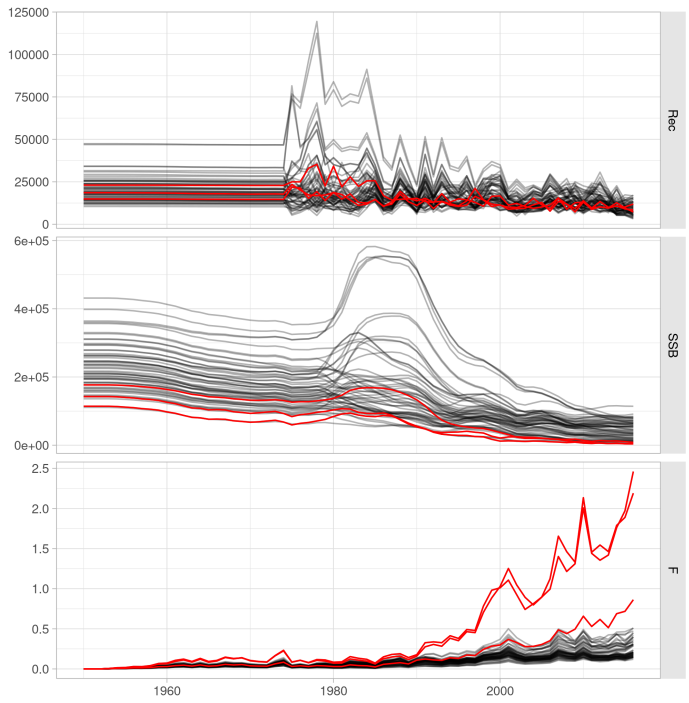


Grid corners

- From M=0.3, sigmaR=0.4, h=0.7, NW CPUE, LF lambda = 0.001 and LL q = 0%
- To M=0.4, sigmaR=0.8, h=0.9, SW CPUE, LF lambda = 1 and LL q = 1%
- 64 (2^6) model runs:
  - 5 with final gradient > 1e-4.
  - 2 with $B_0 > 1e7$.

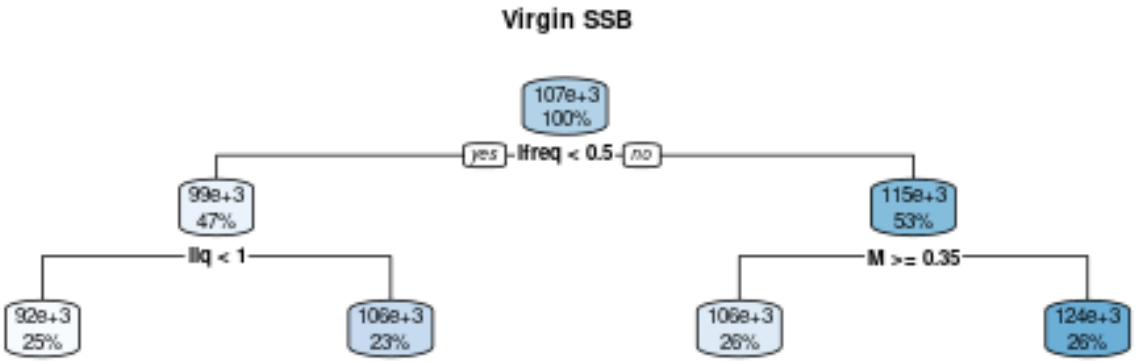


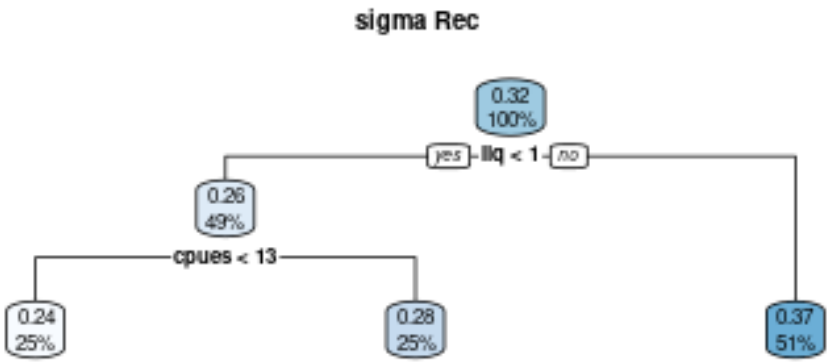
Grid corners

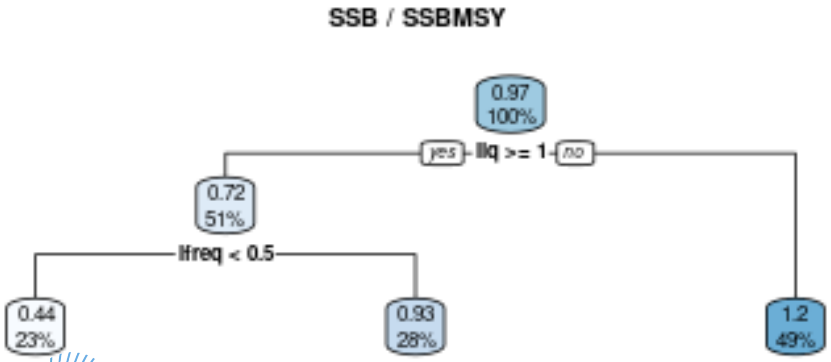


M	sigmaR	steepness	cpues	lfreq	llq
0.3	0.8	0.9	14	1.000	1.00
0.3	0.8	0.7	12	0.001	1.01
0.4	0.8	0.7	12	0.001	1.01

Regression tree







Operating Model reference case grid (108 runs)

- M: 0.3, 0.35, or 0.4, for all ages.
- sigmaR: 0.4, 0.6, or 0.8.
- h: 0.7, 0.8 or 0.9.
- lfreq: 0.01, or 1. Maybe others.
- llq: 0% or 1% per year.
- LL CPUE serie Southwest (14).

Alternatives

- LL CPUE Northwest (12).
- lfreq weighting of PS LF.

Open questions

- Smaller grid still sufficient ?
- Weighting scheme worth considering or . . .
- . . . Equal weighting and resampling ?
- Choice of reference case CPUE (SW 14).
- Range of ages F calculation: 1-12